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Alzheimer's Disease: A Suitable Case for Treatment with Precision Medicine?

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Highlights of the Study

- The heterogeneous and multifactorial nature of Alzheimer's disease needs new therapeutic approaches.
- Precision medicine aims to tailor treatment to specific patients or patient subgroups.
- Various tailored approaches, especially genomic editing techniques, represent a promising new therapy for Alzheimer's disease.
- It is highly likely that precision medicine is most efficacious at the preclinical stage of Alzheimer's disease.

Keywords
Alzheimer's disease · Therapy · Small molecules · Antibodies · Antioxidants · Gene editing

Abstract
Alzheimer's disease (AD) is the most common cause of neurodegenerative impairment in elderly people. Clinical characteristics include short-term memory loss, confusion, hallucination, agitation, and behavioral disturbance. Owing to evolving research in biomarkers, AD can be discovered at early onset, but the disease is currently considered a continuum, which suggests that pharmacotherapy is most efficacious in the preclinical phase, possibly 15–20 years before discernible onset. Present developments in AD therapy aim to respond to this understanding and go beyond the drug families that relieve clinical symptoms. Another important factor in this development is the emergence of precision medicine that aims to tailor treatment to specific patients or patient subgroups. This relatively new platform would categorize AD patients on the basis of parameters like clinical aspects, brain imaging, genetic profiling, clinical genetics, and epidemiological factors. This review enlarges on recent progress in the design and clinical use of antisense molecules, antibodies, antioxidants, small molecules, and gene editing to stop AD progress and possibly reverse the disease on the basis of relevant biomarkers.

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Introduction

Observational studies have led to the understanding that the heterogeneous and multifactorial nature of AD needs an approach other than what is currently used [1,

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2]. Taking this into account, precision medicine (PM) aims to tailor treatment to specific patients or patient subgroups. This relatively new platform would categorize AD patients using a holistic approach on the basis of parameters like clinical aspects, brain imaging, genetic profiling, clinical genetics, and epidemiological factors like environment and lifestyle [3]. Furthermore, it is likely that this framework will allow further evaluation of the effectiveness of newly developed drugs in well-defined groups of patients. To illustrate this, recent research in oncology has identified a drug that can inhibit special functional domains of a protein that is involved in cancer progression [4]. Such a PM-related finding is just an example of how scientists may find the right molecule to target and stop a detrimental process by individual therapy. On the neurodegeneration front and driven by high-tech biotechnology [5], recent research uses three-dimensional models designed to mimic the human brain. This development has issued guidance to further PM-oriented research into therapy of this high-impact disorder [6, 7]. In the subsequent sections, we will elucidate the ways precision targeting may help modify the AD risk and dwell on inflammatory oxidative stress as well as genetic causes. To this end, we will also elaborate the recent therapeutic strategies based on the genomics and pharmacological research, but first, we will highlight the essential role of the apolipoprotein E (ApoE) protein in the pathogenesis of AD.

Amid the expansion of current research on AD, the developing landscape around apolipoprotein E is of extreme importance as its polymorphism is considered the strongest genetic risk and determinant [8–10]. Genome-wide studies have found thousands of causal mutations for human disease, but under the intriguing title "The complex genetic architecture of Alzheimer's disease..." Andrews et al. [11] mention that until 2023, just 90 independent variants across 75 AD/dementia susceptibility loci have been discovered. Among these, numerous studies have determined that the genotype ApoE4 is strongly associated with sporadic late-onset Alzheimer's disease (LOAD) [12]. The human ApoE, located at 19q13.2, is primarily expressed in three isoforms (ApoE2, ApoE3, and ApoE4), all secreted by astrocytes. Frieden and Garai [13] were among the first to describe the ApoE protein in more detail at position 112. The most important protein that is associated with the high risk factor for AD is ApoE4, whereas ApoE3 is only slightly active and ApoE2 is protective. All isoforms have a mass of 34.15 kDa and contain 299 amino acids, but ApoE 3 and ApoE4 contain an arginine residue and ApoE 2 as well as ApoE3 a cysteine residue [14]. Concerning heredity, twin analysis of the total genetic contribution of ApoE4 revealed that this explains about 9% of cases [15]. The LOAD risk increases by 3–4 fold by carrying one copy and 10–15 fold for those carrying two copies of ApoE4, compared to ApoE3 carriers [16, 17]. The different effects among the isoforms are ascribed to a diminished clearance of Aβ by ApoE4, due to decreased expression of enzymatic degradation of the fibrils by neprilysin [18]. Furthermore, mutually divergent molecular structures, affecting binding and clearance, presumably account for the higher risk of Aβ deposits in ApoE4 carriers [19]. The effect of ApoE4 accounts for about 27% of the estimated disease heritability of about 80% [20]. Mechanistically, ApoE is involved in brain lipid homeostasis, a process in which free fatty acids and cholesterol move, through lipidated ApoE as a carrier, to neurons, where lipids are removed from their carrier protein. In neuronal cultures and animal studies, lipid dyshomeostasis causes increased production of Aβ peptides as well as augmented levels of phosphorylated tau [21–23]. Furthermore, detailed cell studies on the N-terminal of APOE and neuronal viability have suggested that interaction with mitochondrial function leads to inflammation as well as Aβ deposition and tau pathology [24, 25]. Altogether, excessive presence of ApoE4 affects lipid transport, Aβ clearance, and synaptic function, which appear to affect women more than men [26–29]. It appears that the ApoE isoforms are not directly associated with tau seeding and propagation in the brain [28, 29]. Regarding specific targeting, it is assumed that the ApoE4 protein is not just a risk factor but can open pathways to initiate AD. Therefore, the targeting of ApoE4 holds promise to improve AD symptoms and, on the longer run, diminish AD occurrence in (sub)groups [29]. Below, we elucidate various techniques that are currently explored to target bioactive AD risk factors.

Antisense Oligonucleotides

Antisense therapy uses antisense oligonucleotides (ASOs) to alter mRNA in order to modulate gene expression and regulate aberrant protein production [30]. Methodological advances in the design of ASOs allow the targeting of brain mRNA and create a new class of therapeutics for AD on the horizon. More specifically, the targeting of short ASOs, founded on Watson-Crick base pairing, can offer the high specificity needed to safeguard delivery to the selected site of action in the brain. The various strategies for mRNA regulation by ASOs have been reviewed by Grabowska-Pyrzewicz et al. [31]. Among the

recently identified targets, Roy et al. [32] have explained that microRNA650 (miR-650) is altered in the brains of various AD patients. This miR-650 targets APOE, presenilin, and cyclin-dependent kinase 5 (CPK5), which play a pertinent role in the pathogenesis of AD. In a mouse model, Lu et al. [33] have confirmed that manipulation of CPK5 by miR-650, based on ASO design, results in reduced plaques and the presence of Aβ in the brain. A phase 1 randomized clinical trial on the antisense oligonucleotide compound B11B080 in 102 patients with mild AD demonstrated that this exploratory drug did reduce tau biomarkers, including CSF t-tau, CSF p-tau181, and tau PET, and corresponded with stable or improved cognitive function. Effects of BI-IB080 on biomarkers and clinical outcomes are being further evaluated in a phase 2 trial [34] under ClinicalTrials.gov Identifier: NCT05399888, ending in December 2026 [34, 35]. To the best of our knowledge, this drug under investigation is the first that prevents tau-mRNA from turning into protein. At the initial stage of this trial, the dose-dependent study was carried out in different groups, and CSF total tau and p-181 tau were reduced by 60 percent in the higher-dose cohorts and stayed low for 6 months. Interestingly, over this period, the reduction was consistently seen in all or most people in each dose group. No serious side effects were reported. Mild adverse events were more common during drug use and included nausea, vomiting, diarrhea, fatigue, confusion, and musculoskeletal pain. There were three cases of mild tinnitus in treated patients and none in the placebo group. This phase 1 human study came after groundbreaking results obtained in a mouse model of tauopathy in which reduction of tau RNA was reported by DeVos et al. [36].

Small Molecules

Small molecules have the potential to penetrate the blood-brain barrier and target detrimental enzymatic action. Considering the complexity of the 3D structure of RNA with its toxic repeats that are often hidden in molecular foldings, innovative drug design is necessary. Toward this goal and to increase the possibility of an efficacious engagement of small molecules with 3D-RNA, medicinal chemistry, including X-ray crystallography, NMR spectroscopy, cryo-electron microscopy, and computational molecular technology, has been shown to be indispensable [37, 38]. New drug designs would increase the probability of beneficial molecular interaction between small molecules and their target [39].

The essential role of BACE1 (beta-site of APP-cleaving enzyme 1) has been recognized as a target to modulate the formation of this neurotoxic entity in the brain [39]. To this end, different small molecules have been synthesized by Ghosh and Oswal [40]. An informative review by Yao et al. [41] lists possible BACE1 inhibitors, but despite reduction of Aβ in the brain, CSF, and plasma in preclinical studies, none of the selected compounds has entered a phase III trial on grounds of serious adverse effects or lack of clinical benefit [42]. A different approach has been the development of small-molecule inhibitors of gamma-secretase, an enzyme that is essentially involved in the biochemical processing of Alzheimer's amyloid precursor (APP) for the generation of Aβ peptides which are responsible for the generation of plaques. Among this type of inhibitors, only semagacestat has entered phase III clinical trials but was terminated due to severe side effects such as cleavage exhibition of other substrates and aggravation of cognitive deficits [43]. Other gamma-secretase modulators with promising preclinical properties have not obtained commercial status for routine use in AD patients (for a few considerations on trial failures, see ref [44]).

The apparent failures in Alzheimer drug research have brought forward a change in the design of lead molecules, as has been suggested by Childs-Disney et al. [45], which comes down to the development of drugs that bind to multiple structures simultaneously within the targeted RNA. Already about 10 years ago, this concept has taken root in essential research based on the function of the neurotransmitter acetylcholine (ACh) in AD patients [46]. The loss of cholinergic neurons in AD patients due to the enzymatic degradation of this molecule by acetylcholinesterase (AChE) into choline and acetate has long been considered responsible for cognitive decline. Currently, the targeting of this enzyme is successfully achieved by drugs such as donepezil, rivastigmine, and tacrine that function as enzyme inhibitors, although the progression of AD is not stopped or reversed by these drugs despite the enhanced concentration of AChE in the synaptic cleft. Present research on tacrine derivatives shows beneficial effects in preclinical studies. These drugs can simultaneously modulate AChE and the tau-related glycogen synthase 3β and show both inhibition of the degrading enzyme and diminished Aβ self-aggregation [47]. This means that new avenues are opened, especially as favorable intestinal absorption and blood-brain permeability have been predicted [48].

Antibodies

In recent years, various clinical trials have employed monoclonal antibodies for targeting Aβ. In this context, the amyloid cascade hypothesis has served as the lead,

although this model may be less applicable to sporadic AD [49]. Detailed information on antibody targeting and its efficacy is available [50]. It has been mentioned that the Clinical Dementia Rating Scale in particular shows improvements for bapineuzumab treatment in cognition and function for mild or moderate AD patients. However, in 2014, Salloway et al. [51] published the results of two phase 3 trials in which no clinical effect of the treatment in mild to moderate AD cases could be demonstrated versus placebo, although increases in plasma Aβ suggested that it was cleared from the brain. Their two study groups involved 1121 APOE4 carriers and 1,331 non-carriers. At week 78, the between-group differences in the change from baseline in the ADAS-cog11 and DAD scores (bapineuzumab group minus placebo group) were −0.2 ($p = 0.80$) and −1.2 ($p = 0.34$), respectively.

In contrast to this antibody, two other anti-amyloid monoclonal antibodies named lecanemab (Leqembi®) and aducanumab (Aduhelm®) did get FDA approval for the treatment of Alzheimer's disease, although the statistical evidence has not been overwhelming [52]. This treatment option is an emerging issue and is illustrated by the recent FDA approval in 2024 of lecanemab-irmb. In a multicenter, double-blind phase III study using this antibody in 898 patients and placebo in 897 patients, this drug appeared to slow the rate of cognitive decline by about 27% over a period of 18 months in early-stage AD patients. In 4 measures of cognition and function, patients receiving the drug compared favorably to placebo-receiving participants ($p < 0.001$) [53]. In addition, the same study group, Cohen et al. [54], concluded that "Lecanemab was associated with a relative preservation of the Quality of Life and less increase in caregiver burden, with consistent benefits seen across different quality of life scales and within scale subdomains. These benefits provide valuable patient-reported outcomes. Indeed, previously reported benefits of lecanemab across multiple measures of cognition, function, disease progression, and biomarkers demonstrate that lecanemab treatment may offer meaningful benefits to patients, care partners, and society" [54].

In this connection, it should be mentioned that disease-modifying antibodies, showing diminished clinical decline, are exceptional in the sense that they intervene in the basic pathophysiologic process. On a further interesting note, it has been demonstrated that the effects of the controversial anti-amyloid antibody aducanumab can be improved by a local low dose of ultrasound that would, at least partly, open the blood-brain barrier and increase its delivery to targeted brain areas of interest [55]. These developments can pave the way for further monoclonal antibody research, although, as of yet, the side effects of these molecules may hamper widespread use [50].

CRISPR/CAS9 Genome Editing

In conjunction with the need for more personalized medicine, the CRISPR/Cas9 system, as a gene processing method, has advanced the biomedical field considerably. It makes it possible to selectively modify DNA sequences at defined sites. In short, a guide DNA targets a gene of interest, and together with the protein Cas9, containing a protease, a double-strand break is caused. If a sister chromatid is present, homology-directed repair can occur with a gene knock-in, and precise genomic change can be achieved. If DNA ends are joined without any homology, a repair mechanism occurs that ultimately results in gene knock-down. A recent review by Nojadeh et al. [56] delivers an overview of the biotechnical know-how and the possible applications of the CRISPR/Cas9 technology in the laboratory environment and in possible clinical applications. In what is a major step forward for gene therapy, the delivery of the edited genome into the mammalian cell preferably takes place with an adeno-associated virus. This virus easily penetrates the host cell and can infect various types of tissues. Moreover, the virus complex, based on diverse capsids, can stay active for longer than 1 year, independent of changing pH and temperature. This Nobel Prize winning development made space for a new wave of other approaches to gene editing. These include methods in which the altered gene can enter the host cell utilizing a plasmid [57] as well as a method that allows cell-specific targeting through selected guide RNA [58]. In familial AD, dominant autosomal presenilin-1 (PSEN1) mutations are responsible for the process of Aβ 42, rather than Aβ 40, aggregation in the brain. A study by Konstantinidis et al. [59] and comments by Thompson [60] described a promising application of CRISPR/Cas9 editing by disrupting the PSEN1M146L allele in peripheral cells of mutation carriers in more than 50% of the treated samples. This procedure reversed the abnormal Aβ 42/40 ratio, which opens the way to a therapeutic strategy in early onset AD patients. Their results are in line with results obtained by Guyon et al. [61] who modified the amyloid precursor protein (APP) gene, that is accountable for plaque formation resulting from excessive cleavage of APP by beta-secretase. Interestingly, this APP A673T mutation can reduce β-secretase cleavage by 40% and is associated with neuroprotection as was found in an Icelandic elder

population. This suggests that the introduction of the APP mutation in the neurons of AD patients could be beneficial. Indeed, their experiments in mixed cultured cell lines showed a considerable reduction in the accumulation of Aβ 42 (60%) and Aβ 40 (81%).

The field of bioethics has long attempted to deal with the possible unintended consequences of emerging technologies. While genome editing has become indispensable and of high value for present and future scenarios, it is necessary to make sure it serves the public. Selin et al. [62] advocate investigating different communities, their needs, and their wants. They recommend the input of experts from biological sciences, law, policy, private industry, and private organizations to weigh, balance, and evaluate the evolution of human genome editing.

Antioxidants

The imbalance between free radicals (reactive oxygen and nitrogen species) on one side and scavengers on the other side in an organism can play an important part in disease mechanisms. Free radicals originate from endogenous sources such as phagocytic cells and peroxisomes as well as from exogenous sources such as pollution and radiation [63]. Oxidative stress supposedly also plays a large part in AD as many components of the neuronal cell are vulnerable to oxidation. Here, damage to highly oxidizable lipids and proteins by hydroxyl radicals in the brain promote Aβ deposition, tau-hyperphosphorylation, and subsequent loss of neuronal function. Under such conditions, the onset of inflammation may occur through the release of pro-inflammatory cytokines such as tumor necrosis factor (TNF)-α, interleukin (IL)-1β, IL-6, and IL-10. The presence of Aβ deposits causes further release of these immune agents by microglia and astrocytes, which causes a vicious circle soon after the onset of AD, eventually leading to tau hyperphosphorylation and excessive neuronal death. Further evidence that inflammation is involved in the onset of AD has recently been provided by Li et al. [64] who demonstrated that hydrocortisone mitigates the generation of reactive oxygen species in a mouse model; this underscores the possible therapeutic use of this anti-inflammatory molecule. As a side note, the suggestion to use anti-inflammatory agents as a treatment for AD was made by 1,992 by McGeer and Rogers [65]. In this context, Rani et al. [66] found an interesting spike in cytokines after induction of AD in mice. The researchers demonstrated that the enhanced concentration of AD-related cytokines was found not only in brain tissue but also in serum and in hepatic and renal tissue. This suggests that pro-inflammatory cytokines can serve as biomarkers, probably even at an early stage of AD. This is important as this disease may be present 15–20 years before recognizable symptoms are observed, emphasizing the start of therapy at an early stage [67].

Appropriate nutritional habits are generally considered to prevent disease owing to the presence of antioxidants [68]. Components that protect against ROS include selenium, zinc, and vitamins C and E, and various molecules, including the promising molecule resveratrol, have been studied extensively. Unfortunately, there is no clear information on the blood concentration of resveratrol after intake, which can be caused by confounding factors such as the intake of water and the presence of plasma proteins, uric acid, and antioxidant enzymes [69]. Moreover, recent research by Zhang et al. [70] found no sufficient evidence that antioxidant imbalance has a significant causal effect on neurodegenerative diseases such as AD, Parkinson's disease, and amyotrophic lateral sclerosis, which casts doubt on the significance of the imbalance hypothesis. Of course, this does not mean that the Mediterranean diet with a mixture of ingredients such as fresh fruit, legumes, vegetables, nuts, unrefined carbohydrates, and starches, does not have an impact on the onset of AD. Herein also plays the gut-brain axis an important role as the gut-microbiome evidently influences cognitive brain function in AD. A poor diet together with external lifestyle factors is held responsible for neurodegeneration, as has been suggested by a recent review of 125 studies over the years 2011–2021 by Puri et al. [71].

Concluding Remarks

As mentioned above, there is scientific agreement that the APOE4 gene is a prominent factor in the genesis of AD. Over the past years, several attempts have been made on potential therapeutic targets for APOE4. Regarding specific targeting, it is assumed that the APOE4 gene is not just a risk factor but can open pathways to initiate AD, and it is therefore plausible that the molecule itself should be the focus of research. Against this background, now that genome editing with the CRISPR technique has advanced, the conversion of the culprit APOE4 gene into the APOE3 or the APOE2 gene would be a viable approach to diminish the Aβ load. Here, the alteration of the amino acids Arg-61 and/or Glu-255 would effectively reduce the risk of AD [72, 73]. Among other interesting

developments, it is worth mentioning that a non-specified lead molecule developed by Kantor et al. [74], produced by a dedicated CRISPR editing strategy to reduce APOE4, demonstrated a remarkable reduction of APOE4 in both a humanized mouse model and in miniature brains derived from a patient with AD, without altering levels of APOE 2 and APOE3. In 2019, Anzalone et al. [75] published on the "prime genome editing" technique, which does not use double strand breaks or donor DNA, that may lead to chromosomal abnormalities. This method offers precise gene editing in the field of genomic substitution, insertion, or deletion for the correction of pathogenic alleles [76]. Following this evolution, numerous applications have been suggested, but the in vivo delivery of the biological product remains a challenge. A possible solution for this problem has been suggested by the results of experiments by An et al. [77], who used engineered virus-like particles to deliver prime editor ribonucleoprotein to restore protein expression for partial visual rescue in mouse models. These exciting developments coincide with a special interest in employing artificial intelligence in AD [78, 79]. The application of advanced hardware and software can handle a multitude of complex patient data on an integrative and time-related progression of AD. Within the scope of this article, the identification of cognitive decline will be supported and untwined by means of the database provided by artificial intelligence. Ideally, this database will relate clinical observations to changes in, for instance, biochemical markers, structural and functional brain regions as detected by MRI and PET as well as lifestyle characteristics [80]. The integration of these data derived from cohorts and trials is definitely of importance to detect AD at early onset in individual patients and facilitate the decision regarding the starting point of treatment. Also in this field, further research will discover fresh biomarkers and relevant biochemical pathways. This is undoubtedly an arduous process that requires the efforts of worldwide research and faces the necessity of working in a collaborative mode. Given the progress that has been made over the past 10 years, it brings with it a promise that the next decade will see a breakthrough. To reach this goal, a method to detect this impairment by biomarkers of synapse dysfunction at its earliest stage is indispensable (see, for instance, ref. [81]).

Statement of Ethics

This article is based on published papers and thus does not violate any principles of ethics.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

Ernest K.J. Pauwels conceived the first and subsequent drafts. Gerard J. Boer provided written input on the scientific content of drafts. Both authors approved the final draft for submission.

Data Availability Statement

This review has used data, independently obtained and agreed upon by both authors, from public sources, including PubMed, Google, and MEDLINE.

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